

Assignment: RNN-Based Text Classification Using AG News Dataset

Building and Comparing RNN, LSTM, GRU, and Bi-LSTM Models for News Topic Classification

Assignment Title

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Dataset

Use the AG News Topic Classification Dataset, a publicly available dataset for news classification.

Install the required package first:

```
!pip install datasets
```

Then load the dataset:

```
from datasets import load_dataset

dataset = load_dataset("ag_news")
```

The dataset contains news articles classified into 4 categories:

Label	Category
0	World
1	Sports
2	Business
3	Sci/Tech

Problem Statement

You are required to build deep learning models that can classify news articles into one of four categories: World, Sports, Business, or Sci/Tech.

In this assignment, you will preprocess text data, convert text into sequences, apply padding, build RNN-based models, and compare the performance of Simple RNN, LSTM, GRU, and Bi-LSTM models.

Learning Objectives

1. Understand how sequence data is represented in NLP.
2. Tokenize and pad textual data for deep learning models.
3. Build a Simple RNN model for text classification.
4. Build and compare LSTM and GRU models.
5. Implement a Bidirectional LSTM model.
6. Apply gradient clipping to control exploding gradients.
7. Compare different RNN-based architectures using evaluation metrics.

Assignment Tasks

Task 1: Load and Explore the Dataset

Load the AG News dataset using the code provided above.

Perform the following:

1. Display the dataset structure.
2. Print the number of training and test samples.
3. Display 5 sample news articles with their labels.
4. Check the distribution of all 4 classes.
5. Plot the class distribution using Matplotlib.

Expected Output:

- Dataset structure
- Number of training samples
- Number of test samples
- Five sample news articles with labels
- Class distribution table
- Bar chart showing the number of samples in each category

Task 2: Text Preprocessing

Prepare the news text for model building.

Perform the following:

1. Extract the text and label columns.
2. Tokenize the text using a suitable tokenizer.
3. Limit the vocabulary size to 20,000 words.
4. Convert text into numerical sequences.
5. Apply padding to make all sequences the same length.
6. Use a fixed maximum sequence length such as 100 or 150.
7. Prepare labels in the required format for multi-class classification.

Expected Output:

- Shape of processed training data
- Shape of processed test data
- Shape of training labels
- Shape of test labels
- Brief explanation of why tokenization and padding are needed

Task 3: Build a Simple RNN Model

Build a Simple RNN model for news topic classification. The model should include an Embedding layer, Simple RNN layer, Dense hidden layer, and an output layer with 4 neurons and softmax activation. Train the model and evaluate it on the test dataset.

Expected Output:

- Model summary
- Training accuracy
- Validation accuracy
- Test accuracy
- Training loss and validation loss curve
- Training accuracy and validation accuracy curve
- Short observation on model performance

Task 4: Build an LSTM Model

Build an LSTM-based model for AG News classification. The model should include an Embedding layer, LSTM layer, Dense hidden layer, and an output layer with softmax activation. Train and evaluate the model.

Expected Output:

- Model summary
- Training accuracy
- Validation accuracy
- Test accuracy
- Loss curve
- Accuracy curve
- Comparison with the Simple RNN model

Task 5: Build a GRU Model

Build a GRU-based model for news classification. The model should include an Embedding layer, GRU layer, Dense hidden layer, and an output layer with softmax activation. Train and evaluate the model.

Expected Output:

- Model summary
- Training accuracy
- Validation accuracy
- Test accuracy
- Loss curve
- Accuracy curve
- Comparison with Simple RNN and LSTM models

Task 6: Build a Bidirectional LSTM Model

Build a Bidirectional LSTM model for news topic classification. The model should include an Embedding layer, Bidirectional LSTM layer, Dense hidden layer, and an output layer with softmax activation. Train and evaluate the model.

Expected Output:

- Model summary
- Training accuracy
- Validation accuracy
- Test accuracy
- Loss curve
- Accuracy curve
- Comparison with regular LSTM
- Short explanation of whether using both forward and backward context improved performance

Task 7: Apply Gradient Clipping

Train one RNN-based model again using gradient clipping.

Perform the following:

1. Select any one RNN-based model from the assignment.
2. Train the selected model without gradient clipping.
3. Train the same model with gradient clipping.
4. Compare training behavior before and after gradient clipping.

Expected Output:

- Model trained without gradient clipping
- Model trained with gradient clipping
- Training loss curve before gradient clipping
- Training loss curve after gradient clipping
- Short observation on whether gradient clipping made training more stable

Task 8: Compare All RNN-Based Models

Create a comparison table for all models.

Model	Training Accuracy	Validation Accuracy	Test Accuracy	Training Time	Key Observation
Simple RNN					
LSTM					
GRU					
Bi-LSTM					

Write a short conclusion covering:

1. Which model performed best?
2. Did LSTM or GRU perform better than Simple RNN?
3. Did Bi-LSTM improve classification performance?
4. Which model trained faster?
5. Which model would you recommend for this dataset and why?

Final Submission Requirements

Submit one Jupyter Notebook containing:

1. Dataset loading and exploration
2. Text preprocessing
3. Tokenization and padding
4. Simple RNN model
5. LSTM model
6. GRU model
7. Bi-LSTM model
8. Gradient clipping implementation
9. Model comparison table
10. Final conclusion

Evaluation Rubric

Criteria	Marks
Dataset loading and exploration	10
Text preprocessing, tokenization, and padding	15
Simple RNN implementation	15
LSTM implementation	15
GRU implementation	15
Bi-LSTM implementation	15
Gradient clipping implementation	10
Model comparison and interpretation	10
Code readability and notebook presentation	5
Total	110

Final Conclusion Prompt for Learners

At the end of the notebook, write a conclusion answering:

1. Which model achieved the best test accuracy?
2. How did Simple RNN compare with LSTM and GRU?
3. Did Bi-LSTM improve classification performance?

4. Did gradient clipping make training more stable?
5. Which model would you choose for this dataset and why?